Design decisions in Ingesting and analyzing data of a global retail

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The case study:
global retail


Global
Chain
Walmart
of Locautions
SUBWAY
Hundands
Locations
of Nunberds
Locations
of Thousrads
of Locations
SUBWAY
Hundads
of Thousands
Thousands
of Thousaons

10,000s of stores worldwide

- **7-Eleven**: One of the largest convenience store chains in the world, with more than **83,000** locations across 19 countries. It's particularly prominent in Japan, the United States, and various parts of Asia.
- **Subway**: The fast-food chain has over **37,000** restaurants worldwide, although it has closed some stores in recent years.
- **McDonald’s**: While not a traditional retail chain, McDonald’s operates more than **40,000** restaurants in over 100 countries.
- **Walmart**: It operates around **10,500** stores globally, including its subsidiaries like Sam's Club and stores in international markets like Mexico and Canada.

100s-1,000,000s distinct SKUs

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- **7-Eleven**: On average, a single store carries **2,500 - 3,000 SKUs**, but this number can vary depending on location and local preferences
- **Subway:** It manages **200 - 300 SKUs** across its entire menu and supply chain
- **McDonald's**: Similarly, a typical location handles **200 to 300 SKUs**.
- **Walmart**: A typical Walmart in the U.S. carries **120,000-150,000 SKUs** across various categories (groceries, electronics, clothing, and home goods). It offers **millions of SKUs** online.

The objective

Ingest, analyze, and support decisions

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- **Ingest** sales data from a global retail, i.e., **each** single **bill** **and** **restock**
- Keep **real-time inventory** (what the **stock of each SKU** across all stores) **for** supporting supply chain decisions (e.g., **restoking** given store’s **SKUs** from warehouses)
- Perform **real-time price analysis** to support store managers' decisions **to promote SKUs** (e.g., those for which there is an overstock)

Designing …
(at high level)

Event-based Systems

Data Stream Mng. Sys.

event sourcing

write once

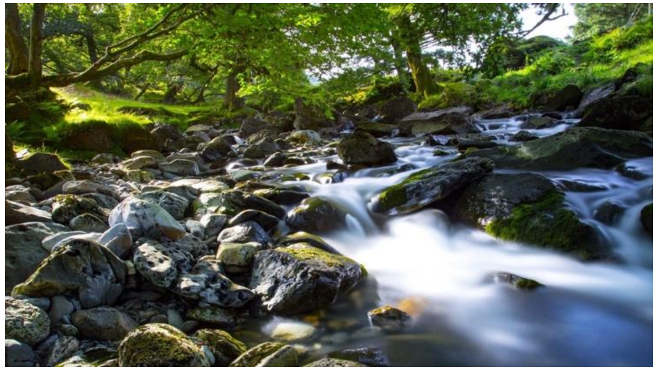
Topic centric communication

Event Store

read many times

event consumers

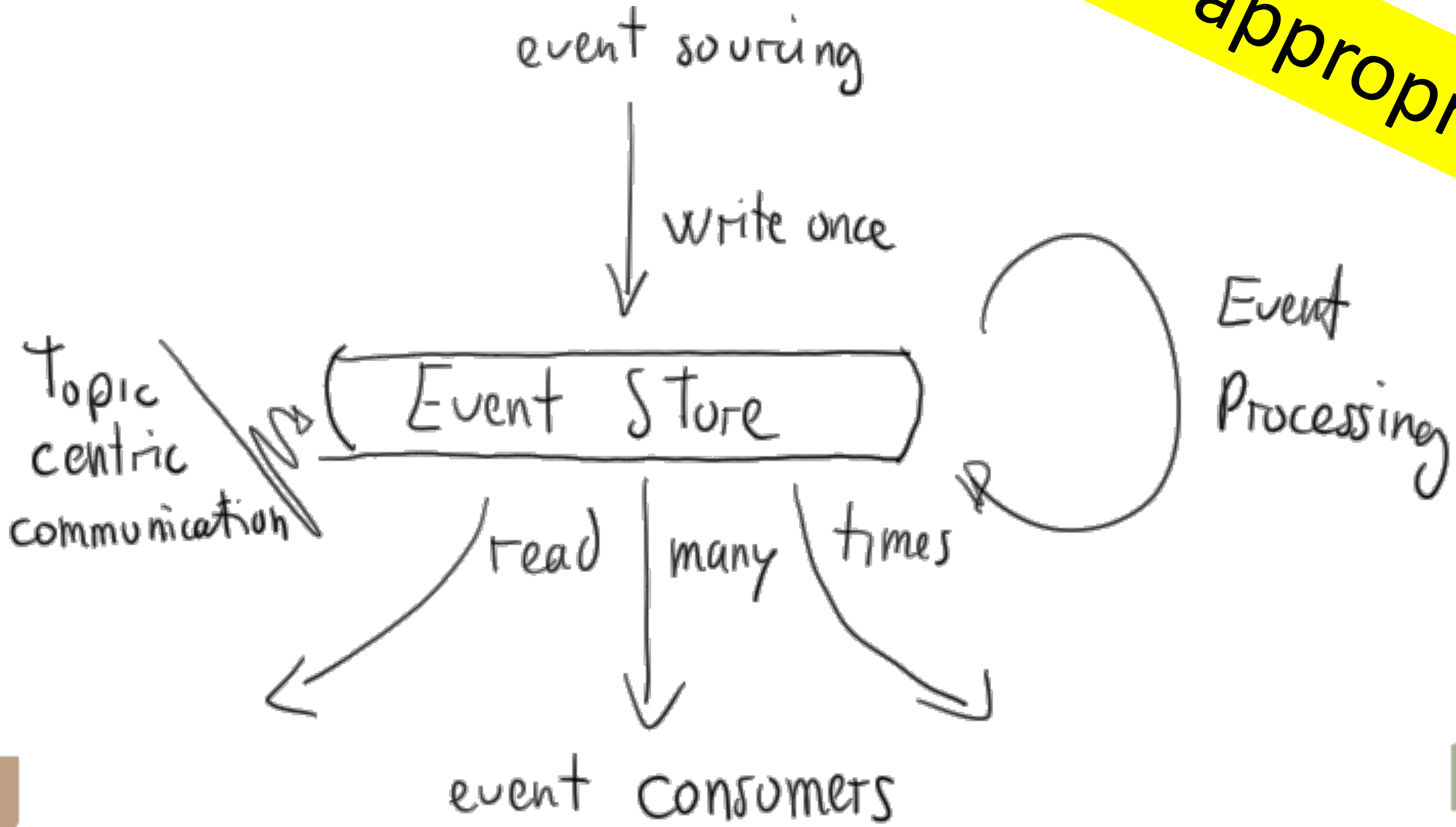
Event Processing



Complex Event Proc.

RECALL

Color the appropriate parts



support decisions

Q: What are the roles of kafka?

Correct

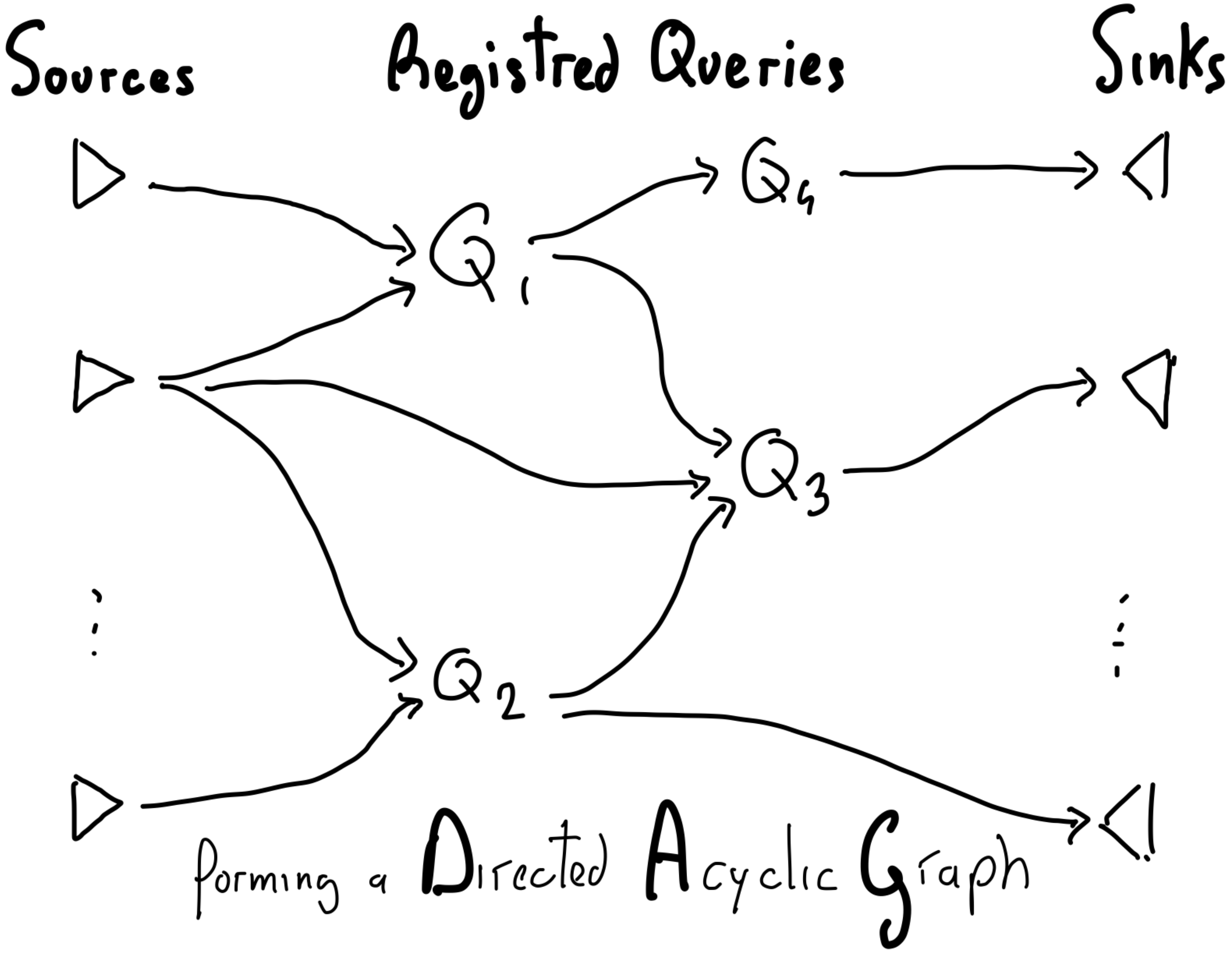
Relevant

Q: What are the roles of Spark?

Correct

Relevant

Typically queries are placed in a network



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Sources

Registred Queries

Sinks

bills

restocks

store inventory t_{-1}

restocking recommendations

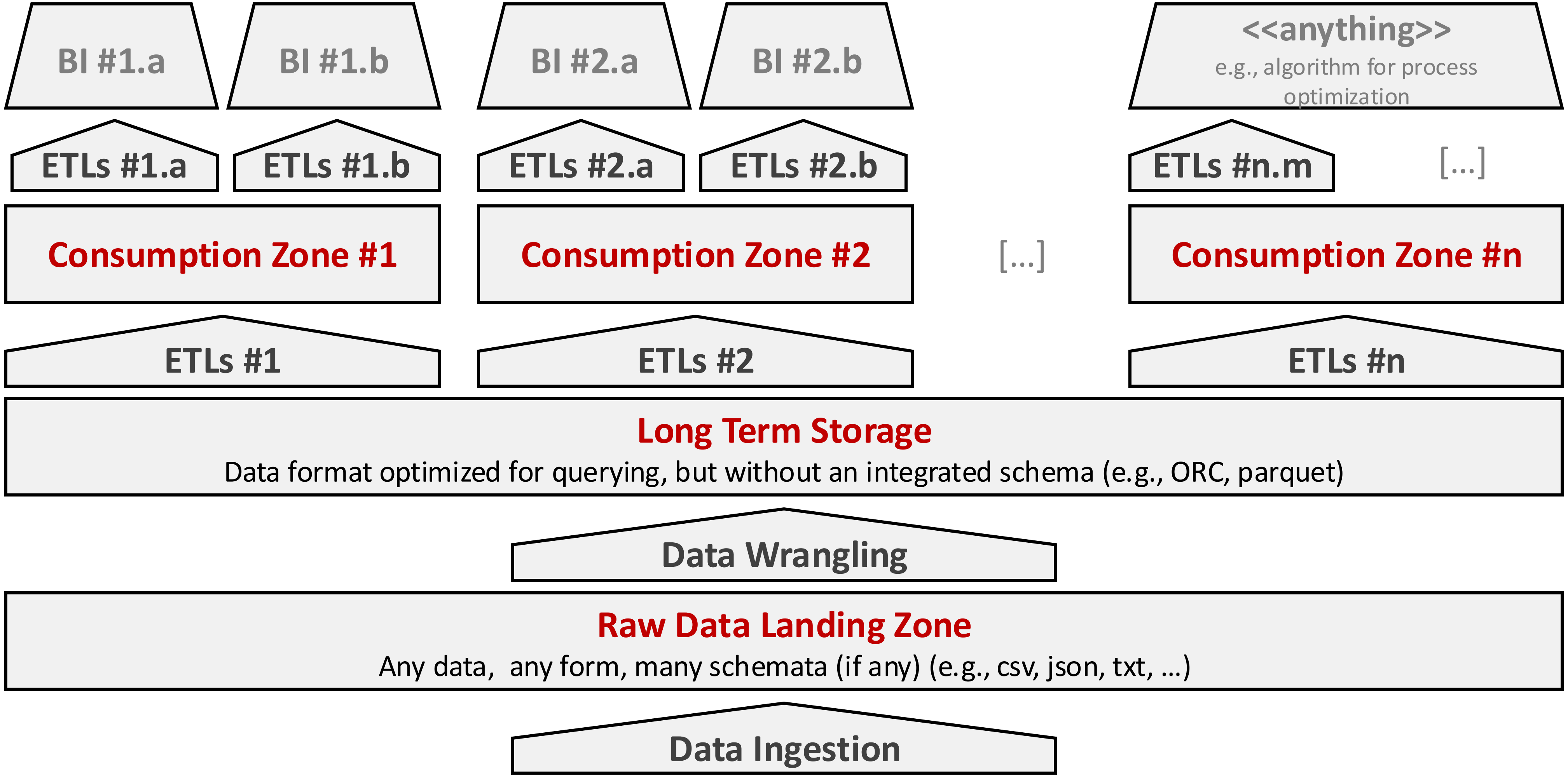
promotions recommendations

store inventory t_0

Place here
the continuous queries
in a DAG

$Q_{b.agg}$ $Q_{r.agg}$ Q_{deltas} Q_{join} $Q_{restocks}$ $Q_{promotions}$

Logical Architecture of a Data Platform



Designing …
(at low level)

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APACHE kafka’s design dimensions (template)

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- For each message
  - Key: null | simple value | structured value
  - Value: simple value | structured value
  - Typical dimension: <1KB – 1 MB
- Producers
  - Number: at need
  - Where: at need
  - How often: at need
- Brokers
  - Number: at need
  - Where: distributed | centralized | hybrid
- For each topic
  - Partition criteria: round robin | hash-based
  - Number of partitions: as many as you need
  - Number of replicas: as many as you need
  - Retention policy: time-based | size-based
  - Clean-up policy: delete old | compaction
- Consumer Groups
  - Number: at need
  - Number of consumers in the group: at need
  - Where: at need

Q: Which are Bills topic's design decisions?

- Message
  - Key:
  - Value:
  - Typical dimension:
- Producers
  - Number:
  - Where:
  - How often:
- Brokers
  - Number:
  - Where:
- Topic
  - Partition criteria:
  - Number of partitions:
  - Number of replicas:
  - Retention policy:
  - Clean-up policy:
- Consumer Groups
  - Number:
  - Number of consumers in the group:
  - Where:

Q: Which are Restock topic's design decisions?

- Message
  - Key:
  - Value:
  - Typical dimension:
- Producers
  - Number:
  - Where:
  - How often:
- Brokers
  - Number:
  - Where:
- Topic
  - Partition criteria:
  - Number of partitions:
  - Number of replicas:
  - Retention policy:
  - Clean-up policy:
- Consumer Groups
  - Number:
  - Number of consumers in the group:
  - Where:

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And so long
and so forth
for all other topics …

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